

4. (a) Describe the advantages of household ring main circuits.

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(b) Explain which safety device operates when the following faults occur.

(i) The ring main circuit stops working when an additional fault-free appliance is plugged into it. [2]

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(ii) The circuit stops working when a lawnmower accidentally cuts the power cable during use. [2]

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(c) Explain why electrical appliances with a metal casing require an earth lead whereas those with a plastic casing do not. [2]

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